



MS Dhoni's first look

MS Dhoni's first look as Atharva in mythological sci-fi graphic novel *Atharva: The Origin* revealed. <https://digit.in/mar22-61>



Oldest known hybrid animal

The mysterious kunga, part donkey, part wild ass were first bred in Syria 4,500 years ago. <https://digit.in/mar22-62>

and other immune system related proteins. Once the desired mRNA is known, the creation process is relatively easy – even faster than the traditional process for vaccines. A computer lays out the sequence of the mRNA, after which a DNA template that can synthesize the aforementioned sequence follows. Very similar to software coding. Plus, another benefit is ending our reliance on animal/cell culture processes involved in vaccine creation – growing it in cells, or chicken eggs.

BEYOND COVID-19

Partnering with the non-profit International AIDS Vaccine Initiative (IAVI), Moderna announced recently that it had entered Phase I of its mRNA vaccine trials against HIV, with the first participants having been successfully dosed.

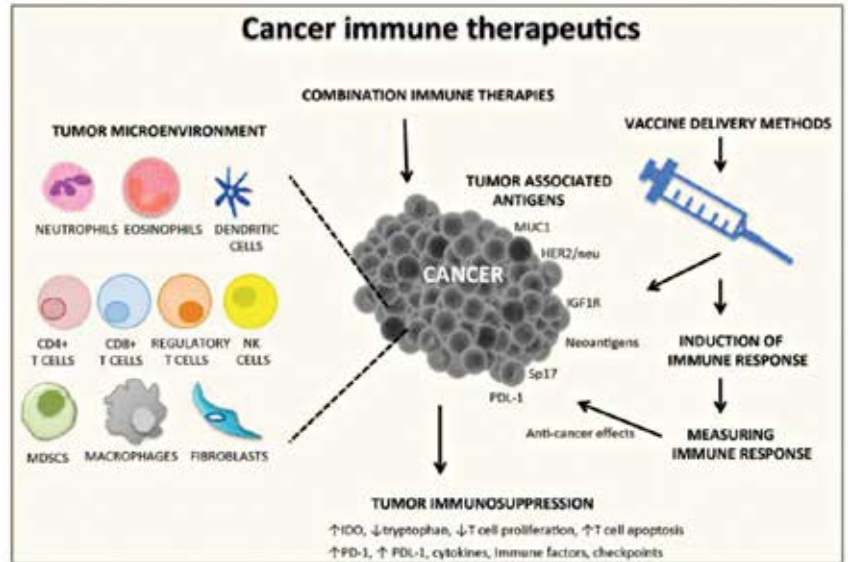
It has been more than four decades since HIV was first discovered, and though successful therapies currently exist to make living with HIV manageable, a vaccine against it is an elusive goal.

The current trial will use Moderna's mRNA tech to provide instructions for immunogen creation, which the cells will then utilize to create the protein. Researchers are also testing an additional boosting immunogen to see if it can assist in generating the desired immune response.

The goal would be stimulating the development of B cells that can produce antibodies that target and neutralize a large range of HIV variants. HIV shows immense genetic diversity, which is a major hurdle that vaccine development will need to tackle. Also,



Could a vaccine for HIV be on the horizon?



Potential vaccine-driven strategies against cancer.

HIV intertwines with the human genome within 72 hours of infection, at which point it is an irreversible, life-long infection.

A REALM OF POSSIBILITIES

Moderna has dozens of prospective mRNA vaccines and therapies in its current pipeline, ranging from vaccines against the Zika and Nipah viruses, to even potential treatments for cancer. BioNTech too, has an equally large repertoire of mRNA projects in the works for various cancers, the flu, and even tuberculosis.

Even notoriously difficult pathogens to target – such as the ones that cause Dengue, infecting hundreds of millions of people every year, especially those in the developing world – could potentially be worked on using mRNA vaccines. Dengue consists of a group of four different viruses (DENV-1/2/3/4) that cause similar disease. Targeting all four dengue viruses is very important, as a subsequent infection with a different Dengue virus can actually end up being more severe. It is thus imperative to neutralize all of them with just one vaccine, which is no easy feat.

A cure for cancer has been a collective human goal for decades. There are two ways in which mRNA can be immensely helpful in anti-cancer

treatment. The first way would be by introducing a person's own antigens to the immune system – ones that are usually shut down in healthy cells, but expressed by the cancer in question. An anti-cancer vaccine would therefore, induce an attack on the cells carrying these antigens at this point. This could be extremely effective, albeit in specific cases – for instance, four antigens can cover up to 90% of melanoma patients, and BioNTech has such a multi-valent vaccine in clinical trials as well!

The second way is even more ambitious – but one that the adaptability of the mRNA technology is particularly well suited to. The vaccines could also target the mutations that occur in the cancer. Though mutations are often personalized to a particular patient, by analyzing the profile, multi-mutation vaccines could be a tangible reality.

A MATTER OF TIME

The future of mRNA tech has far more than the current pandemic in its sights. The future of healthcare itself might rest on the success of mRNA and similar medical technologies. After all, hacking into our own body to reprogram our immune systems to be able to fight illness, new and old, is the stuff science-fiction used to fantasize about. What a time to be alive! [d](#)